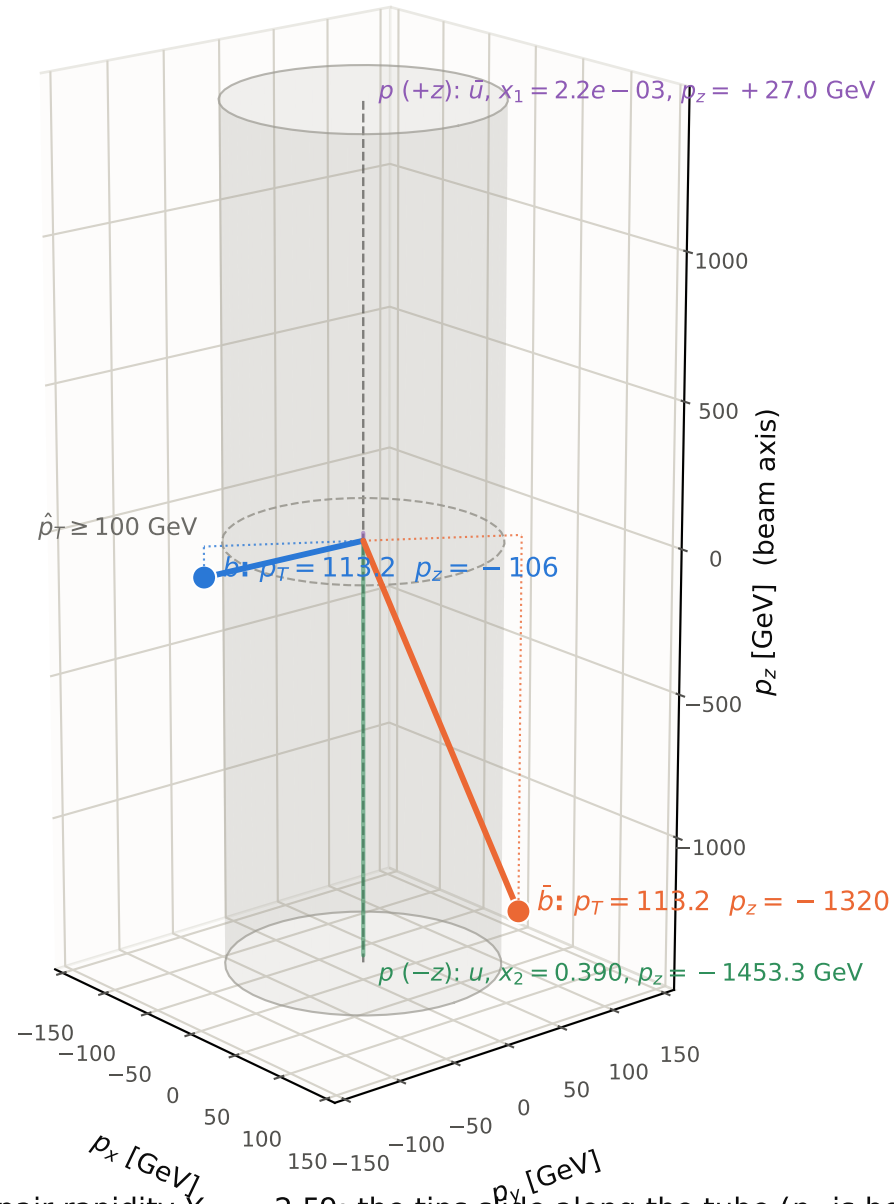


The hard process in three dimensions: partonic CM frame, then the lab frame

scene 1/4



```
event_ME.lhe (process 124)

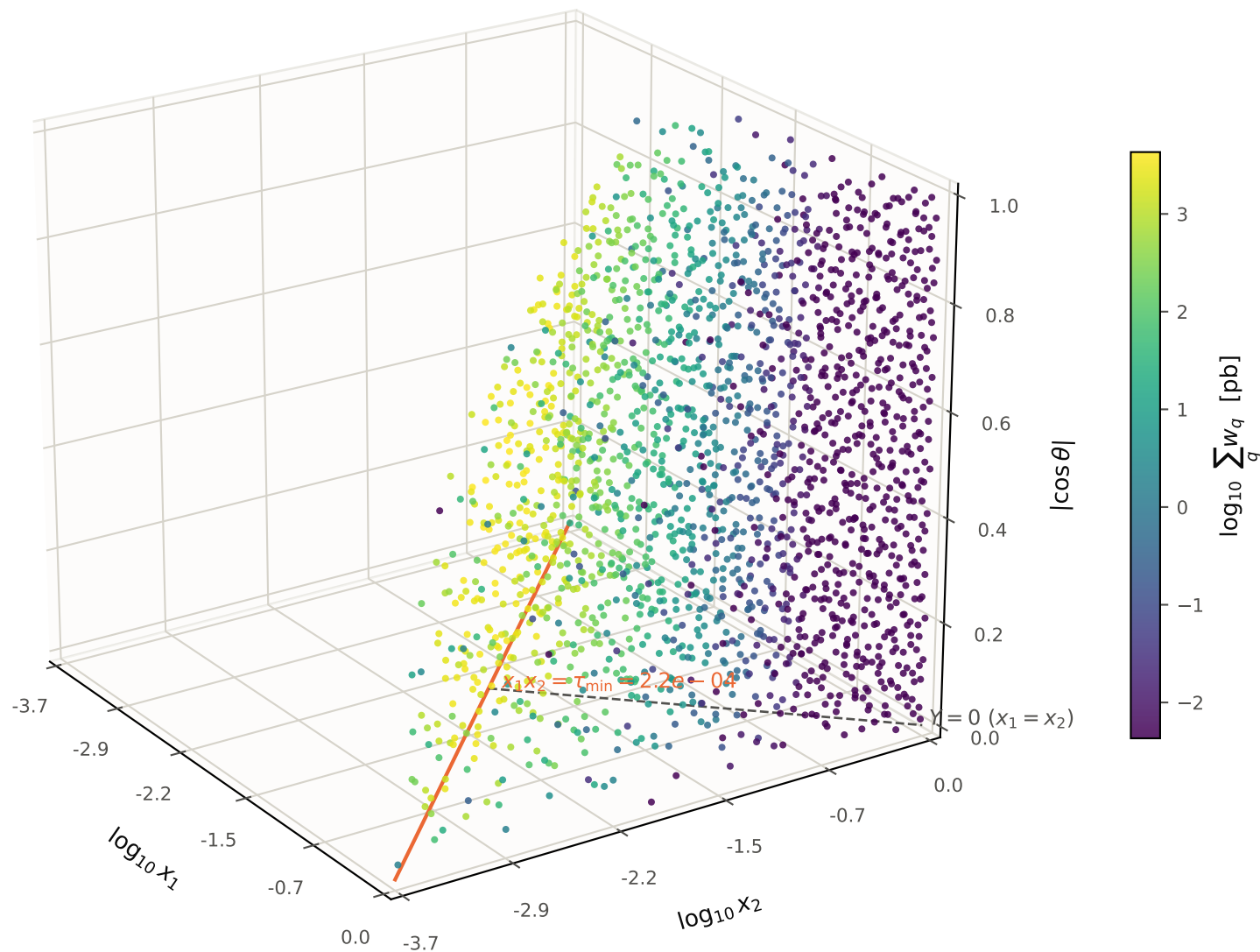
ubar from +z  x1 = 2.174e-03
u    from -z  x2 = 0.3901
sqrt(shat) = sqrt(x1 x2 s) = 396.1 GeV
tau = shat/s = 8.48e-04
p* = sqrt(shat/4-mb^2) = 198.0 GeV
cos theta* = +0.820
pT = p* sin theta* = 113.22 GeV
mT = sqrt(pT^2+mb^2) = 113.32 GeV
pair rapidity Y = -2.59

boost so far = -1.99
y(b) = -0.84    pz = -106.1
y(bbar) = -3.15    pz = -1320.0
pz = mT sinh(y), pT unchanged
```

Boost along the beam by the pair rapidity $Y = -2.59$: the tips slide along the tube (p_T is boost invariant, $p_z = m_T \sinh y$ is not) and the pair moves far forward.

What is integrated: Sobol points in the unit cube, then in physical variables

scene 2/4



```
Sobol points: 2048 of 2048
seed 24680, scrambled

u1 -> ln tau in [ln tau_min, 0]
    tau_min = 4(mb^2+ptmin^2)/s
              = 2.17e-04
u2 -> Y in [-Ymax, Ymax]
    Ymax = -ln(tau)/2
u3 -> |cos theta| in [0, c_max]
    c_max = sqrt(1 - ptmin^2/p*^2)

x1 = sqrt(tau) e^{+Y}
x2 = sqrt(tau) e^{-Y}

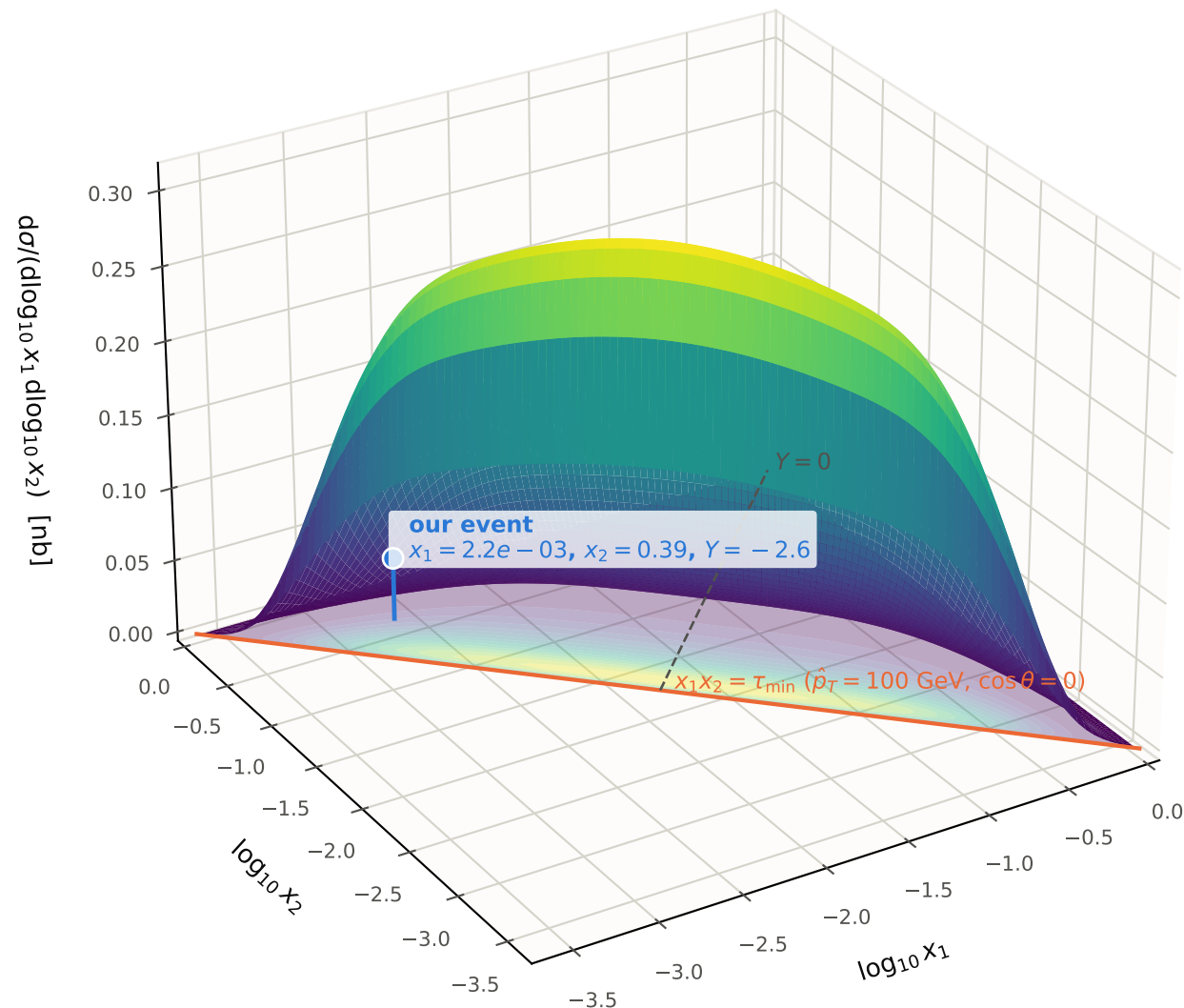
w_q = ln(1/tau_min) 2Ymax 2c_max
      x dsigma/dcos
      x [x1f_q x2f_qbar + (q<->qbar)]

<sum_q w_q> here: 0.345 nb
2^16 points: 0.34532 nb
```

Only the region above the threshold line $x_1 x_2 = \tau_{\min}$ is kinematically allowed; near it $c_{\max} \rightarrow 0$. The stretch of this map is the Jacobian carried in each weight.

The integrand over the (x_1, x_2) plane: its volume is the cross section

scene 3/4



surface height:
integrand with cos theta integrated
= $d\sigma/(d\log_{10} x_1 d\log_{10} x_2)$ [nb]

grid: 120 x 120 cells in (u1,u2)
32 midpoints in u3

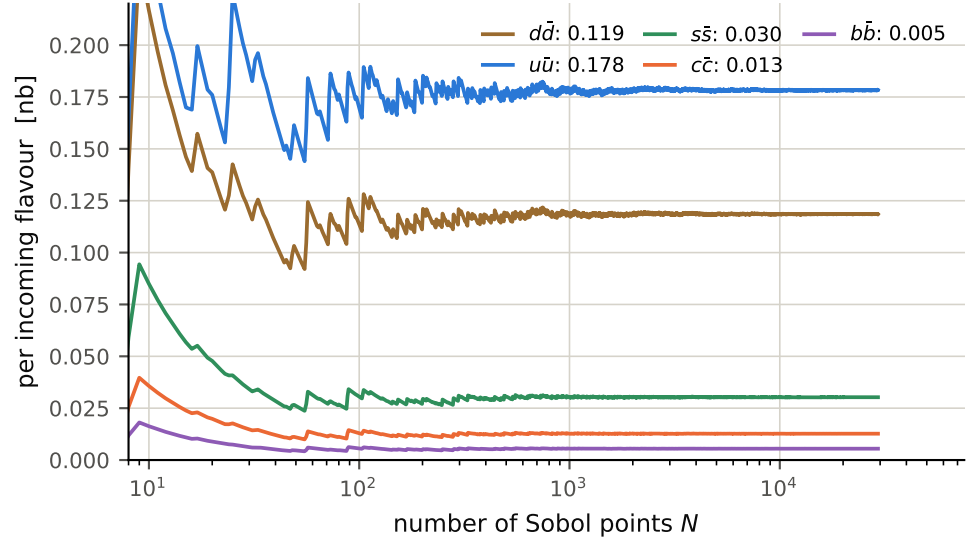
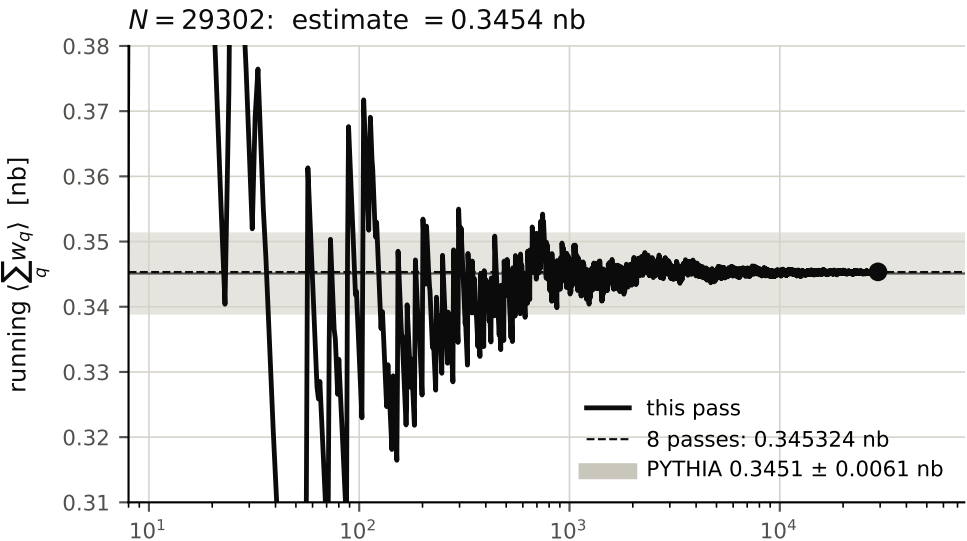
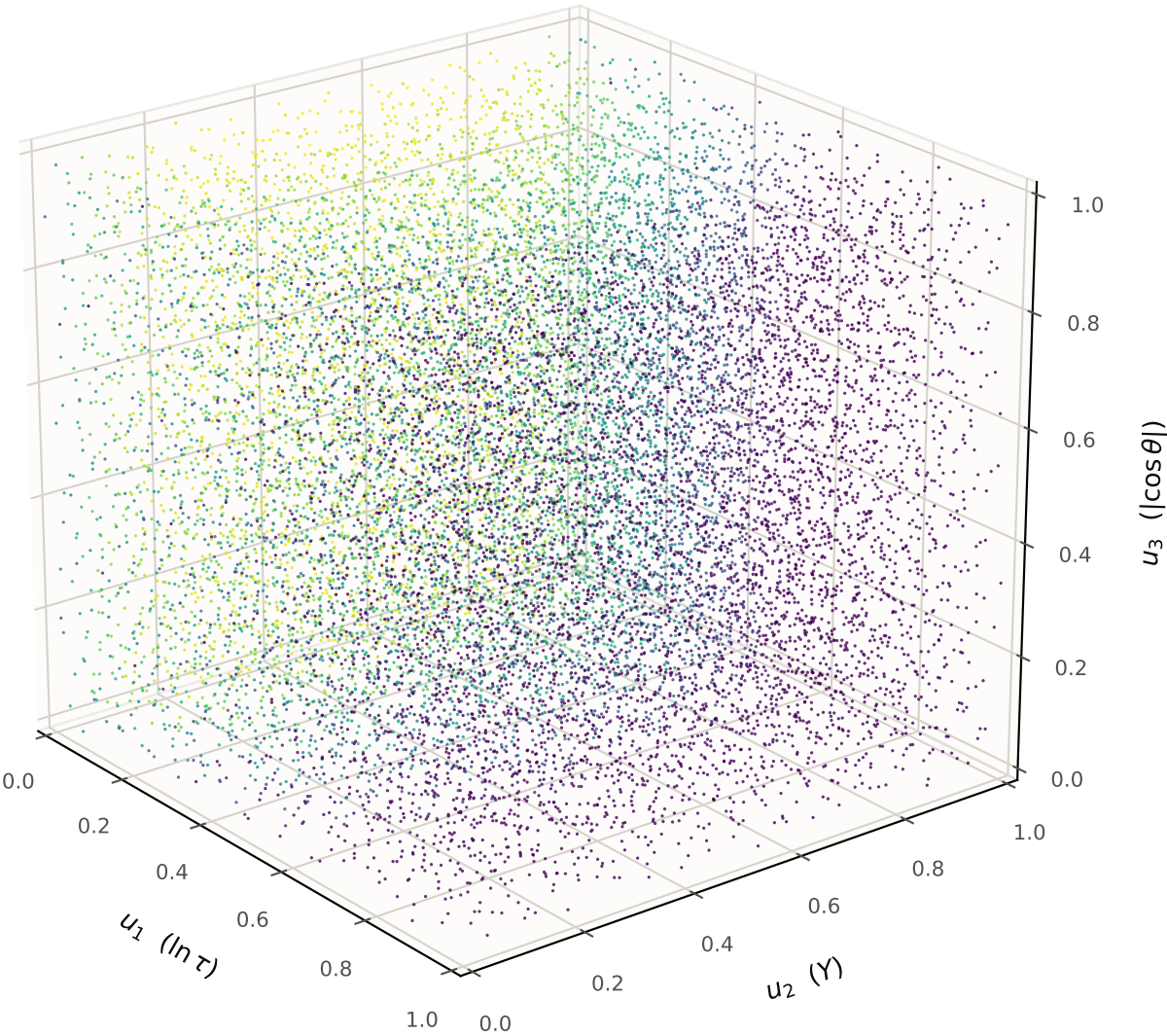
peak 0.2969 nb at
 $x_1 = 1.7e-02$, $x_2 = 1.8e-02$
($Y = 0$, just above threshold)
volume on this grid: 0.3463 nb
→ 0.3453 nb as the grid is refined
Sobol 2¹⁶ (script): 0.34532 nb

front edge: $x_1 x_2 = \tau_{\min}$, $c_{\max} \rightarrow 0$
flanks: $|Y| \rightarrow Y_{\max}$, one $x \rightarrow 1$, PDF $\rightarrow 0$
back: $\tau \rightarrow 1$, PDFs and $1/\hat{s}$ small

The dome sits just above the threshold $x_1 x_2 = \tau_{\min}$ at $Y = 0$: the PDFs grow towards small x and $\hat{\sigma} \propto 1/\hat{s}$, so most of the rate is just above the p_T cut.

Averaging over the cube: one Sobol pass converging to the cross section

first $N = 29302$ Sobol points (16384 of 29302 shown)



After 2^{16} points this pass gives 0.3453 nb; eight scrambled passes give 0.345324 ± 0.00005 nb, PYTHIA's own Monte Carlo 0.3451 ± 0.0061 nb.